

Spring Boot AI Integration Test Document

Application Overview

EduConnect is an educational application that integrates Spring Boot with Ollama's TinyLlama AI model. The application provides document upload capabilities and question-answering features using Retrieval Augmented Generation (RAG).

Technical Stack

- **Backend**: Spring Boot 3.5.6 with Java 17
- **AI Integration**: Spring AI 1.0.0-M6
- **Local AI**: Ollama with TinyLlama model
- **Document Processing**: Apache PDFBox 2.0.30
- **API Documentation**: Springdoc OpenAPI 2.5.0
- **Development**: Visual Studio Code on Windows

Features Tested

1. File Upload - PDF document processing
2. Text Extraction - Using PDFBox library
3. AI Integration - TinyLlama model responses
4. RAG System - Context-aware question answering
5. REST API - Swagger UI documentation

Test Questions

Q: What AI model does this application use?

A: The application uses TinyLlama through Ollama.

Q: How do I upload documents?

A: Use the /api/chat/uploadFile endpoint with PDF files.

Q: What is the purpose of RAG?

A: Retrieval Augmented Generation combines document retrieval with AI generation for accurate, context-aware responses.

Q: Where is the API documentation?

A: Available at <http://localhost:8080/swagger-ui.html>

Configuration Details

- Ollama Base URL: <http://localhost:11434>
- Chat Model: tinyllama:latest
- Embedding Model: nomic-embed-text:latest
- Upload Directory: src/main/resources/static/uploadPdfs

Expected Test Results

1. Successful PDF upload
2. Text extraction from uploaded document
3. Accurate AI responses based on document content
4. Proper error handling for invalid files
5. Working Swagger UI interface